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[Editor Name]
Editor / Handling Editor
Physical Review D
American Physical Society

Dear [Editor Name],

I am pleased to submit for consideration by *Physical Review D* the manuscript:

“Two Structural Obstructions to Galactic Dark Matter in Non-Minimally Coupled
Scalar-Tensor Gravity”

Summary. The paper proves two no-go theorems constraining dark-matter production in scalar-tensor extensions of general relativity with coupling $F(\Psi) = 12 + \Lambda_0 \Psi^2$.

Theorem 1 (G_{eff} ceiling). For any theory in this coupling class, the Solar-System Cassini post-Newtonian bound ($|\gamma_{\text{PPN}} - 1| < 2.3 \times 10^{-5}$) limits $\Lambda_0 < 1.29 \times 10^{-3} M_{\text{Pl}}^{-2}$, giving a maximum effective gravitational coupling $G_{\text{eff}}/G_N \leq 1.018$. Flat galactic rotation curves require $G_{\text{eff}}/G_N \approx 3.1$, which demands $\Lambda_0 \geq 0.154 M_{\text{Pl}}^{-2}$. The two constraints are incompatible by a factor of ~ 119 .

Theorem 2 (shape inversion). For any density-dependent coupling $\beta(\rho_b)$ that is monotonically increasing in baryonic density, the predicted gravitational enhancement $E_{\text{model}}(r)$ tracks the baryon density and decreases with galactic radius, while flat rotation curves require an enhancement that increases with radius. The anti-correlation is geometric and parameter-independent; numerical evaluation on NGC 3198 gives Pearson $r = -0.81$.

Together the theorems close the two principal mechanism channels (effective coupling modification and density-dependent fifth force) for dark matter production in the quadratic non-minimal coupling class. Curvature-memory scalar-tensor gravity (CMSTG) is used as the concrete worked example. The scope and limits of each theorem are characterised precisely.

Significance. Both theorems are derived from general arguments (the Brans–Dicke correspondence and monotonicity of density profiles) rather than CMSTG-specific results, making them applicable to the broader family of non-minimally coupled scalar-tensor theories. Theorem 2 in particular applies to any monotonically increasing chameleon or symmetron coupling without further derivation.

Relation to prior work. This paper is a companion short note to a four-paper CMSTG series submitted to JCAP (Papers I–IV; Paper IV documents the worked examples for both theorems). The theorems are stated as general results here; the JCAP paper uses

them as steps in a broader enumeration of mechanism channels.

Originality. All results are original. The paper has not been submitted elsewhere. Simulation code is publicly archived at https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity.

Suggested reviewers (optional):

[SUGGESTED REVIEWER 1: Name, institution, e-mail, expertise]

[SUGGESTED REVIEWER 2: Name, institution, e-mail, expertise]

[SUGGESTED REVIEWER 3: Name, institution, e-mail, expertise]

I am available for correspondence at the address above.

Yours sincerely,

Christopher R. B. Wilson